Lexington Whalen

whalenlex@gmail.com · lxaw.github.io · linkedin.com/in/lxaw · Google Scholar

Profile

Machine learning researcher at the intersection of frontier AI and U.S.–Japan technology cooperation, specializing in efficient LLM training and inference. Co-developed NVIDIA's first diffusion language model series (Nemotron-Labs-Diffusion, Efficient-DLM), then co-led pretraining and inference optimization for SoftBank's Sarashina LLMs. Now a Research Scientist at NVIDIA Research (DLER), working on efficient intelligence. Publications at top conferences; NSF Graduate Research Fellow.

Education

Georgia Institute of Technology — **Atlanta, GA**
M.S. in Computer Science, Specialization in Artificial Intelligence — *Expected Spring 2027*
NSF Graduate Research Fellow.

University of South Carolina, Honors College — **Columbia, SC**
Accelerated B.S./M.S. in Computer Science — *Jan. 2021 – Aug. 2024*
Outstanding Senior, Class of 2023.

Peer-Reviewed Publications

[1] Y. Fu, **L. Whalen**, A. Garg, C. Wu, M. Khadkevich, N. Oswald, E. Xie, D. Egert, S. Turuvekere Sreenivas, S. Diao, C. Yu, Y. Yu, W. Chen, S. Norouzi, S. Lan, L. Zhu, J. Wang, J. Jiang, M. Mardani, M. Maghoumi, S. Han, A. Jukić, N. Tajbakhsh, J. Kautz, P. Molchanov. "Nemotron-Labs-Diffusion: A Tri-Mode Language Model Unifying Autoregressive, Diffusion, and Self-Speculation Decoding." **EMNLP**, 2026. [NVIDIA]

[2] Y. Fu*, **L. Whalen***, Z. Ye, X. Dong, S. Diao, J. Liu, C. Wu, H. Zhang, E. Xie, S. Han, M. Khadkevich, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Efficient-DLM: From Autoregressive to Diffusion Language Models, and Beyond in Speed." **ICML**, 2026. [NVIDIA]

[3] C. Shih, C. Li, C. Yu, H. Fang, S. Li, W. Hsin, **L. Whalen**, H. Suh, G. Eisenhauer, L. Liu, Y. (Celine) Lin. "NeRArch-Sim: A Unified Simulator for Benchmarking and DSE of Neural Rendering Accelerators." **ISCA**, 2026. [Georgia Tech]

[4] **L. Whalen***, Z. Du*, H. You*, C. Li, S. Li, Y. (Celine) Lin. "Early-Bird Diffusion: Investigating and Leveraging Timestep-Aware Early-Bird Tickets in Diffusion Models for Efficient Training." **CVPR**, 2025. [Georgia Tech]

[5] Z. Ye, Z. Wang, K. Xia, J. Hong, L. Li, **L. Whalen**, C. Wan, Y. Fu, Y. (Celine) Lin, S. Kundu. "LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering." **ACL**, 2025. [Georgia Tech]

[6] A. Murphy, **L. Whalen**, S. Dubinsky, M. Gavin, J. F. Bailyn, J. Ginn. "On 'historical unity' of Russian and Ukrainian: A linguistic perspective on language conflict and change." **LSA**, Apr. 2023. [University of South Carolina]

* Equal contribution

Preprints & Technical Reports

[1] **L. Whalen**, Y. Ito, R. Sakamoto. "KBBQ: A Predictive Noise Law and the Limits of Spectrum Flattening in FP4 Quantization." Technical Report, 2026. [SB Intuitions]

[2] **L. Whalen**, Y. Ito, R. Sakamoto. "Teaching Diffusion to Speculate Left-to-Right." Technical Report, 2026. Under review at TMLR. [SB Intuitions]

Patents

[1] **L. Whalen**, Y. Ito, R. Sakamoto. 「拡散ベースの投機的ドラフトモデルにおける受理長改善アルゴリズム」 Japanese Patent (filed), 2026. [SB Intuitions]
"Algorithm to Improve Acceptance Length of Diffusion-Based Speculative Draft Models."

[2] Y. Fu, **L. Whalen**, X. Dong, S. Diao, C. Wu, E. Xie, S. Han, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Training Framework for Converting Autoregressive Language Models into Faster Diffusion Language Models." U.S. Patent (filed), 2026. [NVIDIA]

Research Experience

NVIDIA Research – DLER Team — **Santa Clara, CA**
Research Scientist — *Nov. 2026 –*

- Working on efficient intelligence.

SoftBank – SB Intuitions — **Tokyo, Japan**
Pretraining & Inference Optimization Research Lead — *Apr. 2026 – Nov. 2026*

- Co-led pretraining and post-training optimization research for the Sarashina series, Japan's largest LLMs, at SoftBank's SB Intuitions R&D Division.
- Led novel quantization efforts, spanning models up to hundreds of billions of parameters.
- Built SB Intuitions' first speculative-decoding inference stack for the Sarashina family; introduced training-time drafter-alignment methods that raise the accepted draft length by 21–76% over a position-uniform baseline.
- Served as a key technical voice in Japanese-language cross-team discussions; partnered with leading Japanese and international technology companies and government agencies to evaluate GPU resources and co-develop long-term compute strategy.

NVIDIA – DLER Team — **Atlanta, GA & Santa Clara, CA**
Efficient Deep Learning Intern & Data Filtering Challenge Lead — *Dec. 2024 – Mar. 2026*
Mentors: Pavlo Molchanov & Yonggan Fu (NVIDIA Research).

- Co-developed Nemotron-Labs-Diffusion (EMNLP 2026), a tri-mode (autoregressive / diffusion / self-speculation) language model family at 3B–14B scale; the 8B model decodes 6× more tokens per forward pass than Qwen3-8B at comparable accuracy (up to 4× higher throughput); released on Hugging Face with ∼1M total downloads.
- Co-first author of Efficient-DLM (ICML 2026), converting pretrained autoregressive models into diffusion LMs; the 8B model reaches +5.4%/+2.7% higher accuracy at 4.5×/2.7× higher throughput than Dream 7B / Qwen3 4B (open-weights on Hugging Face); presented directly to CEO Jensen Huang.
- Helped build the training, evaluation, and inference infrastructure for NVIDIA's first series of diffusion language models, adopted by 5+ internal research teams for vision-language models, embeddings, and latent reasoning.
- Organized NVIDIA's 2025 International Data Filtering Challenge for Training Edge Language Models across 30+ university and industry teams.

Georgia Institute of Technology — **Atlanta, GA**
Graduate Researcher — *Aug. 2024 – Dec. 2025*

- Led a team of 3 Ph.D. and Master's students to cut diffusion-model training time by 2.9–5.8× (up to 10.3× vs. standard train-prune-finetune) while maintaining generation quality; CVPR 2025.
- Algorithm lead for the ARPA-H THOR healthcare initiative (with Stanford, Northwestern, Carnegie Mellon, and MIT): designed resource-efficient ML for body-powered cancer-detection sensors, cutting sensor degradation by over 70% and extending sensor lifetime from hours to days.

Honors & Awards

- SoftBank Group Representative, 2026 Entrance Ceremony — Apr. 2026
- NSF Graduate Research Fellowship ($150,000 value) — Apr. 2024
- Toyo University Exchange Student Representative — Jul. 2023
- University of South Carolina Outstanding Senior, Class of 2023 — Feb. 2023
- U.S. Department of State Critical Language Scholar of Japanese — Jan. 2023
- University of South Carolina Top Scholar ($100,000+ value) — Aug. 2020

Talks

- "速く、小さく、軽く：LLM効率化のベストプラクティス", SoftBank – SB Intuitions — Oct. 2026
  Faster, Smaller, Lighter: Best Practices for LLM Efficiency
- "拡散言語モデル：基本から応用まで", SoftBank – SB Intuitions — Jul. 2026
  Diffusion Language Models: From Fundamentals to Applications

Service & Outreach

- Serve as an informal advisor on U.S.–Japan AI and technology policy, engaging with U.S. Embassy Tokyo and Japanese government stakeholders.

Language Skills

English (Native) · **Japanese** (Native-level fluency, JLPT N1 certified) · **Mandarin** (Intermediate)